Boosting Remote Sensing Change Detection via Hard Region Mining

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***Abstract*—Change detection (CD) is essential for various real-world applications, such as urban management and disaster assessment. Numerous CD methods have been proposed, and considerable results have been achieved recently. However, detecting changes in hard regions, for example, the boundaries of changes, small changes, and irrelevant pseudo-changes caused by background clutter, remains difficult for these methods, since they pay equal attention to all regions in bi-temporal images. This article proposes a novel CD network, termed HRMNet, which provides accurate change maps via hard region mining. Specifically, the HRMNet performs hard region mining from two different perspectives: 1) a cross-layer knowledge review module (CKRM) is introduced to identify hard regions embedded in high-level features and distill temporal change information from low-level features to high-level features, thereby enhancing feature representation capabilities and 2) an online hard region estimation (OHRE) branch is constructed to model pixel-wise hard samples, supervised by the error between predicted change maps and the corresponding ground truth during training. Finally, the hard region-aware features extracted from the OHRE branch and multilevel temporal difference features are aggregated into a unified feature representation to improve the accuracy of CD. Experimental results on three benchmark datasets demonstrate the superior performance of the HRMNet in the CD task. The demo code for our work is publicly available at https://github.com/guanyuezhen/HRMNet**

***Index Terms*—Change detection (CD), cross-layer feature fusion, hard region mining, knowledge review, optical remote sensing images.**

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I. Introduction

CHANGE detection (CD) aims to locate and segment the changes that are most relevant in the bi-temporal images captured at diverse periods in the same surface regions. As an important technique in remote sensing sense understanding, CD has received increased attention and been applied for many real-world applications, such as land-use change detection [1], [2], urban management [3], global resources monitoring [4], and damage assessment [5]. Traditional CD methods primarily employ image transformation [6], [7] and image algebra [8], [9], [10], [11], [12] techniques to identify changes in low- or middle-resolution remote sensing images. Previous methods typically utilize principal component analysis [6] and tasseled cap transformation [7] to enhance change information in bi-temporal images, thereby enabling easy detection of changes. Representative image algebra techniques-based methods, such as image difference [8], image regression [9], [10], [11], [12], and change vector analysis [13], are used to formulate difference images and choose suitable thresholds to identify changes. However, traditional CD methods mainly rely on hand-crafted features [14], which significantly limit the detection accuracy for high-resolution and very high-resolution remote sensing CD due to their low robustness and insufficient semantics. Recently, with the development of deep convolutional neural networks (CNNs), deep learning has made significant progress in many computer vision tasks, such as image super-resolution [15], [16], [17], anomaly detection [18], [19], and image fusion [20]. Motivated by the powerful feature representation capability of CNNs, many deep learning-based CD approaches have been proposed and have achieved superior performance compared with traditional CD methods [21].

Previous research has made extensive efforts to improve the accuracy of CD results mainly from the following aspects [14], [22], [23], [24], [25], [26], [27], [28], [29], [30].

1) *Multilevel Feature Fusion:* As we know, the features extracted from the top layers of backbones are beneficial to locating objects, while those extracted from the bottom layers of backbones are conducive to recovering the

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details of objects [31]. In recent years, many works with various well-designed multilevel feature fusion architectures are proposed to boost CD performance [14], [22], [23], [24], [25], [26], [27], [28], [29], [30], [32]. For example, the long short-term memory (LSTM) and skip connection are combined to obtain more distinguishable multilevel features [33]. A full-scale feature fusion manner is introduced to aggregate multilevel information at diverse feature scales. Afterward, hybrid attention is utilized to capture the long-range context information among bi-temporal features for a more accurate CD. Li et al. [34] proposed a guided refinement model, which first aggregates multilevel features and then exploits the aggregated features to repetitively polish multilevel features, to filter out the irrelevant noise information mixed in multilevel features.

2) *Temporal Difference Extraction:* Effectively capturing the temporal difference information from bi-temporal features also plays a key component in CD. Previous methods usually employ feature concatenation or subtraction operations to realize temporal difference extraction. As demonstrated in [35], the original bi-temporal information can be preserved by the temporal concatenation operation, which stacks bi-temporal features along the dimension of channels in order. The feature subtraction needs an additional hand-crafted distance, for example, ℓ_1 or ℓ_2 distance, to measure the pixel-wise temporal difference. Apart from the above two native temporal difference extraction (TDE) manners, Zheng et al. [35] considered the temporal order and designed a temporal-symmetric transformer, which exploits weight-shard 3-D convolutional layers with different temporal arrangement orders to extract temporal difference information. Lei et al. [36] introduced a difference-enhance module (DEM) based on the feature subtraction measure to select the most significant channels of bi-temporal features with channel attention. Li et al. [34] considered the complementary aspects of feature concatenation or subtraction operation and designed a temporal feature interaction module, in which the change regions of bi-temporal features can be highlighted. Fang et al. [37] proposed a feature interaction module that consists of a set of alternative interaction layers in the feature extractor to perform temporal feature interaction of a set of alternative interaction layers in the feature extractor to perform temporal feature interaction.

3) *Attention Mechanisms:* In recent years, attention mechanisms have received intensive attention, due to their abilities in capturing context information and boosting feature discriminability. Channel attention [38], convolution block attention module [39], and nonlocal [40] are three widely used attention mechanisms in the recently proposed CD method. For example, Zhou et al. [41] utilized the multihead self-attention mechanism to capture the intertemporal information and reduce the imaging difference so that the changes can be detected easily. Wang et al. [42] introduced two attentive feature aggregation schemes to merge the cross-layer features in different processes.

In some real-world applications, it is crucial to accurately determine the changes, particularly in some hard detected regions known as hard samples, for example, the boundaries of changes, small changes, and irrelevant pseudo-changes caused by background clutter. Change boundaries is important to indicate the overall changed areas in some large-scale changes. Small changes are difficult to recognize by human beings, but cannot be ignored. Pseudo-changes often arise from different appearances of the same land covers over diverse periods or are caused by similar appearances to actual changed regions, leading to mis-detection. Previous methods typically treat all changed and unchanged regions equally, which can result in insufficient modeling for the above challenging areas. Consequently, accurately identifying hard regions remains difficult. Recently, several approaches have been proposed to address the detection of hard regions. For instance, a boundary-based model [43] was introduced in one study to enhance sensitivity to change boundaries, but it does not account for other hard regions. In [44], the HASNet employs a foreground-aware optimization loss [45] to focus on foreground examples and hard examples of the background during training to achieve a balanced optimization. This type of manner typically uses score maps to identify hard samples, which are then applied as weights in the loss function to encourage the model to focus on these challenging samples. However, some hard regions that exhibit small gradient updates can be easily overlooked by the model during training [46], restricting the abilities for detection hard regions.

Differently, we propose a novel CD network, termed HRM-Net, to address the detection of changes in hard regions from two perspectives.

1) Each level of features encodes specific temporal change information and consequently contains its unique hard regions, particularly in high-level features due to their lower resolutions. To tackle these hard regions across low-level features, we design a cross-layer knowledge review module (CKRM) utilizing reverse attention and conflict attention. The reverse attention allows the network to refocus on regions that are not adequately segmented in the high-level features, while the conflict attention compels the network to concentrate on conflicting regions between detection results from high- and low-level features. This approach facilitates the distillation of knowledge from low- to high-level features, thereby enhancing the representation capability of temporal difference features.
2) Unlike previous methods that employ weighted loss functions to encourage the model to focus on challenging samples, we directly learn hard region-aware features to direct the model to address these difficult regions. To achieve this, we introduce an online hard region estimation (OHRE) branch, which dynamically estimates pixel-wise hard samples, supervised by the discrepancies between the predicted change maps and the corresponding ground truth during training. Ultimately, the comprehensive knowledge of hard region-aware

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features captured from the online estimation branch is integrated with multilevel temporal difference features for accurate CD.

The main contributions of this work are summarized as follows.

1) We propose the HRMNet, a novel CD network that enhances the accuracy of identifying changes through hard region mining.
2) We introduce a CKRM that measures hard regions within high-level features and distills temporal change information from low-level features, thereby improving the feature representation.
3) An OHRE branch is developed to model pixel-wise hard samples, guided by the discrepancies between predicted change maps and ground truth during training.
4) We aggregate hard region-aware features from the online estimation branch with multilevel temporal difference features, resulting in a comprehensive feature representation that significantly enhances CD accuracy.

This article is organized as follows. Section II offers a brief review of related works in hard region mining strategies. Section III introduces the proposed method, detailing the network architecture, the OHRE branch, and the CKRM. In Section IV, we conduct extensive experiments on three benchmark datasets to evaluate the performance of the proposed HRMNet, accompanied by detailed discussions and model analyses. Finally, Section V concludes the article.

II. Related Works

The purpose of hard sample mining is to enhance the discriminative capability of the deep networks. In recent years, the hard sample mining strategy has progressively become an active research topic in some deep learning-based tasks, such as image segmentation [47], multimodal learning [48], face recognition [49], [50], object detection [51], and person re-identification [52].

The hard sample mining strategy, also known as the hard region mining strategy, focuses on selecting pixels or regions that are particularly challenging to classify accurately. This includes identifying hard negatives and hard positives within an image, a critical approach in the field of computer vision. This approach helps alleviate the imbalance between positive and negative samples, ultimately enhancing model training. For example, focal loss [53] assigns significantly larger weights to hard samples with higher loss values, while giving much smaller weights to easy samples with lower loss values. Gu et al. [47] employed two depth-related metrics, that is, the depth prediction error and the depth-aware segmentation error, to evaluate the hard pixels, from which a weight map is formulated to encourage the model to pay more attention to the hard pixels. Hu et al. [48] presented two robust loss functions to make the deep networks focus on clean samples instead of noisy ones, thereby alleviating the interference of noisy samples. Nie et al. [54] introduced a sample selection strategy based on estimated confidence, encouraging the network to focus more on hard samples. A pixel-adaptive convolution filter is employed to estimate confidence, enhance the upsampling operation, and effectively propagate the high-confidence pixels into less reliable regions [55]. Xiao et al. [56] proposed a method to learn pixel hardness by utilizing the hardness information derived from global and historical loss values, thereby enhancing semantic segmentation performance. Inspired by the significant success of hard region mining strategies across various research domains, we propose incorporating this approach into the CD task to enhance the ability of the model to deliver accurate change results for real-world applications. Previous hard region mining methods typically use score maps to identify hard pixels or regions, which are then applied as weights in the loss function to encourage the model to focus on these challenging pixels or regions. In contrast, we directly aggregate hard region-aware features with multilevel temporal difference features, enabling the model to better recognize changes in difficult regions.

III. Proposed Method

In this section, the overview of the proposed HRMNet is first presented. Next, the structures of the OHRE manner and the CKRM are given in detail. Finally, we illustrate the loss function of the HRMNet.

A. Overview

As shown in Fig. 1, we first utilize a shared feature extractor, which follows an encoder–decoder structure, to extract bi-temporal features. For the encoder, any backbone can be used here, and we utilize a ResNet18 [57] as done in previous methods [3], [34], [58] for a fair comparison. Let a registered pair of images be denoted as $\mathbf{I}^t \in \mathbb{R}^{h\times w\times 3}, t \in \{1,2\}$, where h, w, and t denote the height, width, and temporal order of images, respectively. The multilevel bi-temporal features, denoted as $\mathbf{F}_i^t \in \mathbb{R}^{h/2^{i+1}\times w/2^{i+1}\times c_i}, i \in \{2,3,4,5\}, t \in \{1,2\}$, can be obtained from the four stages of the encoder Enc$(\cdot)$, as

$$\mathbf{F}_i^t = \text{Enc}\left(\mathbf{I}^t\right), i \in \{2,3,4,5\}, t \in \{1,2\}. \tag{1}$$

The decoder is implemented in a U-shaped structure, in which a feature aggregation module (FAM) is used to progressively fuse the multilevel context information from high levels to low levels. In the FAM, features are first processed through convolutional layers for feature transition. They are then concatenated and passed through another convolutional layer for feature fusion. This process is denoted as

$$\begin{aligned}\hat{\mathbf{F}}_i^t &= \text{Conv}_{3\times 3}\left(\mathbf{F}_i^t\right)\\ \hat{\mathbf{F}}_{t-1}^t &= \text{Conv}_{3\times 3}\left(\text{Up}\left(\mathbf{F}_{i-1}^t\right)\right)\\ \mathbf{P}_i^t &= \text{Conv}_{3\times 3}\left(\text{Cat}\left(\hat{\mathbf{F}}_i^t, \hat{\mathbf{F}}_{t-1}^t\right)\right)\end{aligned} \tag{2}$$

where Conv$(\cdot)$, Up$(\cdot)$, and Cat$(\cdot)$ denote the convolutional layer, upsampling, and feature concatenation, respectively. $\hat{\mathbf{F}}_i^t \in \mathbb{R}^{h/2^{i+1}\times w/2^{i+1}\times c_i}$ and $\hat{\mathbf{F}}_{t-1}^t \in \mathbb{R}^{h/2^{i+1}\times w/2^{i+1}\times d}$ are the ith and $(i-1)$th features after feature transition, respectively. With the decoder, we capture multilevel bi-temporal features $\mathbf{P}_i^t \in \mathbb{R}^{h/2^{i+1}\times h/2^{i+1}\times d}, i \in \{2,3,4,5\}, t \in \{1,2\}$ for detecting changes, as

$$\mathbf{P}_i^t = \begin{cases}\mathbf{F}_i^t, & \text{if } i = 5\\ \text{FAM}\left(\mathbf{F}_i^t, \mathbf{F}_i^{t-1}\right), & \text{if } i \in \{2,3,4\}\end{cases} \tag{3}$$

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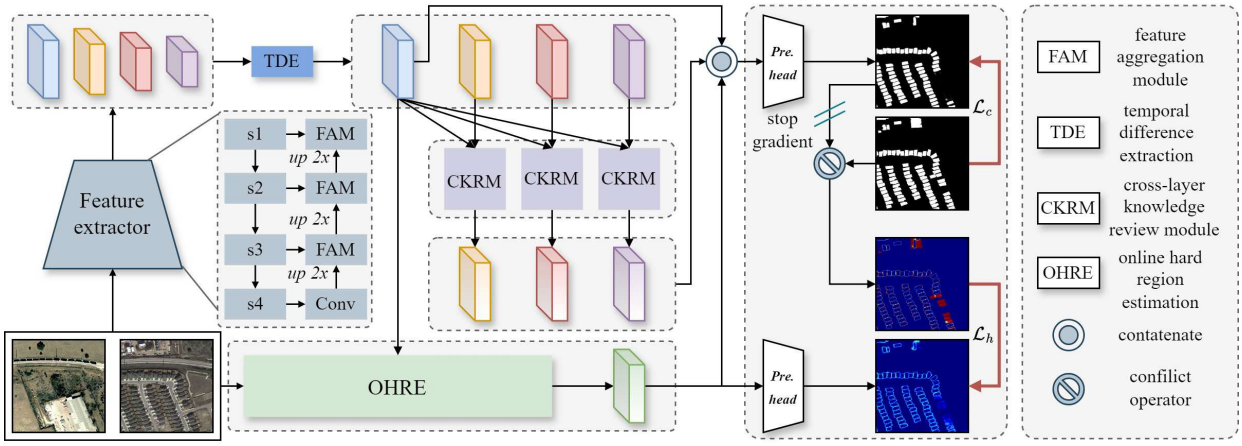


Fig. 1. Framework of the proposed HRMNet. Initially, the bi-temporal images pass through a shared feature extractor to obtain bi-temporal features, and then multilevel temporal difference features are obtained through the TDE. CKRMs fully explore the multilevel temporal difference knowledge to enhance the feature capabilities. The OHRE branch estimates pixel-wise hard samples corresponding to changed and unchanged regions, supervised by the diversity between predicted change maps and corresponding ground truth in the training process. Finally, the multilevel temporal difference features and hard region-aware features obtained from the OHRE branch are aggregated to generate the final change maps.

where FAM(·) denotes the FAM. Subsequently, we employ a feature abstraction and absolution as a TDE module to capture temporal difference information. The procedure of TDE can be represented as

$$\mathbf{D}_i = \mathcal{T}\left(\mathbf{P}_i^1, \mathbf{P}_i^2\right) = \left|\mathbf{P}_i^1 - \mathbf{P}_i^2\right|, i \in \{2,3,4,5\} \tag{4}$$

where $\mathbf{D}_i \in \mathbb{R}^{h/2^{i+1} \times h/2^{i+1} \times d}, i \in \{2,3,4,5\}$ denotes the obtained temporal difference features. $\mathcal{T}(\cdot)$ represents the function of TDE.

To address the hard detected regions, we first employ the CKRM to access the hard detected regions mixed within high-level features, guided by reverse attention and conflict attention. The CKRM further distills temporal change information from low-level features into high-level features for feature refinement. This process can be represented as follows:

$$\hat{\mathbf{D}}_i = \mathrm{CKRM}\left(\mathbf{D}_i, \mathbf{D}_2\right), i \in \{3,4,5\} \tag{5}$$

where CKRM(·) denotes the CKRM. $\hat{\mathbf{D}}_i \in \mathbb{R}^{h/2^{i+1} \times h/2^{i+1} \times d}$ is the ith level of refined features. Besides, we introduce an OHRE branch, which is designed to evaluate pixel-wise hard regions to strengthen the capability of the model to deal with the hard detected changes in bi-temporal images. From the OHRE branch, the hard region-aware features $\mathbf{U} \in \mathbb{R}^{h/4 \times w/4 \times d}$ are further combined with multilevel temporal difference features to make the network pay more attention to the hard regions, as

$$\mathbf{C} = \mathrm{Conv}_{3\times 3}\left(\mathrm{Cat}\left(\mathbf{U}, \mathbf{D}_2, \hat{\mathbf{D}}_3, \hat{\mathbf{D}}_4, \hat{\mathbf{D}}_5\right)\right) \tag{6}$$

where $\mathbf{C} \in \mathbb{R}^{h/4 \times w/4 \times d}$ is the obtained temporal difference features for CD. More details of the cross-layer knowledge review and OHRE will be presented in Sections III-B and III-C, respectively.

B. Cross-Layer Knowledge Review Module

It is commonly acknowledged that the low-level features extracted from the bottom layers of the backbone contain abundant object details, while the high-level features obtained from the top layers of the backbone consist of semantic abstractions about the objects. Thus, different levels of features have their hard detected changes. It is crucial to fully utilize the information in both low- and high-level features to deal with the hard detected changes and achieve accurate CD. To address this issue, we propose a knowledge review strategy consisting of three CKRMs. As illustrated in Fig. 2, a multibranch structure equipped with conflict attention and reverse attention is utilized to capture the complementary information between the temporal difference features $\mathbf{D}_2$ and $\mathbf{D}_3$.

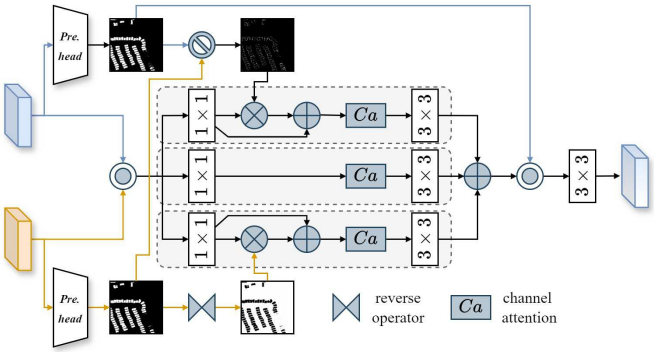


Fig. 2. Illustration of the CKRM. Reverse attention and conflict attention are used together to distill temporal change information from low-level features into high-level features.

The multilevel temporal difference features capture various aspects of changed objects. The conflicts between change maps derived from high- and low-level features can be considered hard regions within the high-level features, as the lower resolution of high-level features captures fewer details about the changes. By distilling change information from low-level features, the hard regions in the high-level features can be addressed to some extent. Let $\mathbf{p}_2^c \in \mathbb{R}^{h/4 \times w/4}$ and $\mathbf{p}_3^c \in \mathbb{R}^{h/4 \times w/4}$ denote the predicted change maps of temporal difference features $\mathbf{D}_2$ and $\mathbf{D}_3$, respectively, the conflict regions can be

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calculated as the difference between $\mathbf{p}_2^c$ and $\mathbf{p}_3^c$

$$\mathrm{CoA} = \mathbf{p}_2^c\left(\mathbf{1} - \mathbf{p}_3^c\right) + \mathbf{p}_3^c\left(\mathbf{1} - \mathbf{p}_2^c\right) \tag{7}$$

where $\mathrm{CoA} \in \mathbb{R}^{h/4\times w/4}$ is the obtained conflict attention between $\mathbf{D}_2$ and $\mathbf{D}_3$.

High-level features are effective for object localization but provide less detail about changed objects, making them more challenging to detect by high-level features. Therefore, we introduce a reverse attention to exploit the low-level feature to compensate for the insufficient detail in high-level features. Specifically, the reverse attention is formulated to erase the changed regions predicted by high-level features, thereby forcing the network to focus more on the unchanged regions. The reverse attention is formulated as

$$\mathrm{ReA} = \mathbf{1} - \mathbf{p}_3^c \tag{8}$$

where $\mathrm{ReA} \in \mathbb{R}^{h/4\times w/4}$ denotes the reverse attention.

Afterward, the CKRM concatenates $\mathbf{D}_2$ and $\mathbf{D}_3$ and passes through three 1×1 convolution layers for feature transition as

$$\begin{aligned}\mathbf{D}_1^k &= \mathrm{Conv}_{1\times1}\left(\mathrm{Cat}\left(\mathbf{D}_2, \mathbf{D}_3\right)\right)\\ \mathbf{D}_2^k &= \mathrm{Conv}_{1\times1}\left(\mathrm{Cat}\left(\mathbf{D}_2, \mathbf{D}_3\right)\right)\\ \mathbf{D}_3^k &= \mathrm{Conv}_{1\times1}\left(\mathrm{Cat}\left(\mathbf{D}_2, \mathbf{D}_3\right)\right)\end{aligned} \tag{9}$$

where $\mathbf{D}_j^k \in \mathbb{R}^{h/4\times w/4\times d}$, $j = 1, 2, 3$, denote the features for three branches of the CKRM. The conflict attention and reverse attention are plugged into two branches, making the network capture the discriminative changed regions. In addition, a channel attention [38] after a 3×3 convolution layer is also injected into each branch of the CKRM for enhancing the feature representation capability. The whole process can be denoted as

$$\begin{aligned}\mathbf{D}_1^r &= \mathrm{Conv}_{3\times3}\left(\mathrm{CA}\left(\mathbf{D}_1^k \oplus \mathbf{D}_1^k \otimes \mathrm{CoA}\right)\right)\\ \mathbf{D}_2^r &= \mathrm{Conv}_{3\times3}\left(\mathrm{CA}\left(\mathbf{D}_2^k \oplus \mathbf{D}_2^k \otimes \mathrm{ReA}\right)\right.\\ \mathbf{D}_3^r &= \mathrm{Conv}_{3\times3}\left(\mathrm{CA}\left(\mathbf{D}_3^k\right)\right)\end{aligned} \tag{10}$$

where $\mathbf{D}_j^r \in \mathbb{R}^{h/4\times w/4\times d}$, $j = 1, 2, 3$, are the enhanced features. $\mathrm{CA}(\cdot)$, $\oplus$, and $\otimes$ represent the channel attention, element-wise addition, and multiplication, respectively. Finally, $\mathbf{D}_1^r$, $\mathbf{D}_2^r$, and $\mathbf{D}_3^r$ are integrated into one representation and further concatenated with the prediction $\mathbf{p}_2^c$ to generate final temporal difference features $\hat{\mathbf{D}}_3$ as

$$\hat{\mathbf{D}}_3 = \mathrm{Conv}_{3\times3}\left(\mathrm{Cat}\left(\mathbf{p}_{2c}, \mathbf{D}_1^r \oplus \mathbf{D}_2^r \oplus \mathbf{D}_3^r\right)\right). \tag{11}$$

C. Online Hard Region Estimation

The goal of hard region estimation is to determine the confidence or hardness related to the prediction of the model on a pixel-wise basis. To achieve this, we introduce an OHRE branch, as shown in Fig. 3(c). OHRE first uses two down-convolution blocks, implemented with three 3×3 convolutional layers, followed by batch normalization and ReLU functions, to extract texture information from the bi-temporal images. The TDE method is then employed to capture temporal difference information, which is further combined with the temporal difference features $\mathbf{D}_2$ via a FAM to capture comprehensive knowledge for hard region estimation. The entire process can be expressed as follows:

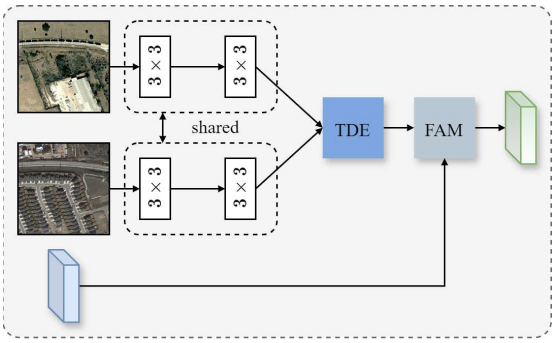


Fig. 3. Illustration of OHRE. Two down-convolution blocks are employed to extract texture information, which is then combined with the temporal difference feature using the FAM to capture comprehensive knowledge for uncertainty estimation.

$$\begin{aligned}\mathbf{T}^t &= \mathrm{DoB}\left(\mathbf{I}^t\right), t \in \{1, 2\}\\ \mathbf{U} &= \mathcal{F}\left(\mathbf{D}_2, \mathcal{T}\left(\mathbf{T}^1, \mathbf{T}^2\right)\right)\end{aligned} \tag{12}$$

where $\mathrm{DoB}(\cdot)$ and $\mathcal{F}(\cdot)$ denote the down-convolution blocks and the FAM, respectively. $\mathbf{T}^t \in \mathbb{R}^{h/4\times w/4\times c_2}, t \in \{1, 2\}$ and $\mathbf{U} \in \mathbb{R}^{h/4\times w/4\times c_2}$ represent the bi-temporal texture features and hard region-aware features, respectively.

Intuitively, the network should have higher confidence in the true-positive and false-negative regions of the prediction, but lower confidence in the false-positive and false-negative regions. Therefore, we aim to use the differences between the prediction of the model and the corresponding ground truth as the supervision signal to guide the learning process of the OHRE branch. The hard regions supervision can be formulated as follows:

$$\mathbf{g}^u = \mathbf{p}_1^c\left(\mathbf{1} - \mathbf{g}^c\right) + \mathbf{g}^c\left(\mathbf{1} - \mathbf{p}_1^c\right) \tag{13}$$

where $\mathbf{g}^u \in \mathbb{R}^{h\times w}$, $\mathbf{p}_1^c \in \mathbb{R}^{h\times w}$, and $\mathbf{g}^c \in \mathbb{R}^{h\times w}$ represent the hard regions supervision, predicted change map, and corresponding true changes, respectively. We train the OHRE branch with a binary cross-entropy (BCE) loss $\mathcal{L}_{\mathrm{bce}}$, which is formulated as

$$\mathcal{L}_u = \mathcal{L}_{\mathrm{bce}}\left(\mathbf{p}^u, \mathbf{g}^u\right) = \mathbf{g}^u \log\left(\mathbf{p}^u\right) + \left(\mathbf{1} - \mathbf{g}^u\right)\log\left(\mathbf{1} - \mathbf{p}^u\right) \tag{14}$$

where $\mathcal{L}_u$ denotes the hard region loss and $\mathbf{p}_u$ is the predicted hard region map.

D. Loss Function

We jointly train the CD part and the OHRE branch. For the CD part, a hybrid loss [34], [59], including a BCE loss $\mathcal{L}_{\mathrm{bce}}$ and a dice (Dice) loss $\mathcal{L}_{\mathrm{dice}}$ [60], is adopted as

$$\mathcal{L}_c = \mathcal{L}_{\mathrm{bce}} + \mathcal{L}_{\mathrm{dice}} \tag{15}$$

where $\mathcal{L}_c$ is the CD loss.

With the hard region supervision, the total training loss of the HRMNet can be formulated as

$$\mathcal{L} = \mathcal{L}_c + \mathcal{L}_u. \tag{16}$$

In summary, we present the complete algorithm for the proposed HRMNet in Algorithm 1.

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Algorithm 1 Boosting Remote Sensing CD via Hard Region Mining

Input: CD model $\phi_\theta(\cdot)$, training dataset $(\mathbf{I}^1, \mathbf{I}^2)$, and maximum training epoch N;
Output: Trained CD model $\phi_\theta(\cdot)$;
for $s = 1, \ldots, N$ **do**
  %% temporal difference feature extraction
  Extract bi-temporal features $(\mathbf{P}_i^1, \mathbf{P}_i^2), i = 2, 3, 4, 5$ through the backbone and FAM in (1), (2) and (3);
  Obtain temporal difference features $\mathbf{D}_i, i = 2, 3, 4, 5$ via (4);
  %% Cross-layer knowledge review
  Generate CD maps $\mathbf{p}_i^c, i = 2, 3, 4, 5$ from $\mathbf{D}_i, i = 2, 3, 4, 5$;
  for $i = 3, 4, 5$ **do**
    Implement conflict attention CoA via Eq. (7);
    Implement reverse attention ReA via Eq. (8);
    Generate refined temporal difference feature $\hat{\mathbf{D}}_i$ via Eqs. (9), (10), and (11);
  end
  %% Online hard region estimation
  Extract hard region aware feature $\mathbf{U}$ via Eq. (12);
  Generate hard region aware map $\mathbf{p}^u$ from $\mathbf{U}$;
  Construct hard regions supervision $\mathbf{g}^u$ via Eq. (13);
  Compute hard region supervision loss $\mathcal{L}_u$ via Eq. (14);
  %% Generate Final Change Map
  Fuse temporal difference features $\mathbf{D}_2$, $\hat{\mathbf{D}}_i, i = 3, 4, 5$, and hard region aware feature $\mathbf{U}$ via Eq. (6);
  Generate final CD map $\mathbf{p}^1$ from $\mathbf{C}$;
  Compute CD loss $\mathcal{L}_c$ for CD maps $\mathbf{p}_i^c, i = 1, 2, 3, 4, 5$ via Eq. (15);
  Obtain total loss $\mathcal{L}$ via Eq. (16);
  %% Back-propagation
  Do back-propagation and update model parameters $\phi_\theta(\cdot)$;
end
Return trained CD model $\phi_\theta(\cdot)$.

IV. Experiments

A. Datasets

In our experiments, we use three remote sensing CD benchmark datasets to evaluate the effectiveness of the proposed HRMNet. The detailed information is listed as follows.

LEVIR+ [61][1]: It is an expanded version of LEVIR [3], which consists of 637 pairs of bi-temporal remote sensing images captured from the Google Earth platform with 1024 × 1024 spatial size and 0.5 m spatial resolution. LEVIR+ includes an additional 348 bi-temporal remote sensing images, with both datasets mainly focusing on building changes. In our experiments, we merge the validation and testing set of LEVIR to form a training set for LEVIR+. The additional 348 bi-temporal remote sensing images are used as a separate testing set. Due to limited GPU memory, all images are cropped into nonoverlapping 512 × 512 patches. A total of 1780/768/1392 image pairs are obtained for training/validation/testing, respectively.

BCDD[2]: It is a building CD dataset containing a 32507 × 15354 bi-temporal images with 0.075 m resolution. In our experiments, we crop the images into nonoverlapping patches of size 512 × 512 and then obtained a total of 1827 pairs of bi-temporal images for the out-domain testing.

MCD [62][3]: It encompasses a wide range of open-pit mining sites located in diverse countries and regions across six continents to examine land-use and land-cover changes resulting from mining operations. For each site, the dataset includes bi-temporal paired images with a spatial resolution of 1.2 m. In our experiment, we randomly choose 60%, 30%, and 10% of mining sites as the training, validation, and testing sets, including a total of 10289, 5404, 1597 bi-temporal 512 × 512 patches, respectively.

B. Evaluation Metrics

To verify the effectiveness of the proposed HRMNet, five widely used evaluation metrics [36], [63], [64], namely the Kappa coefficient (κ), intersection over union (IoU), $F1$-score ($F1$), recall (Rec), and precision (Pre) are adopted in our experiments. The detailed calculations of five metrics can be found in [34] and [59]. Note that a higher value for each metric indicates better CD results and demonstrates a strong ability to identify changes in hard regions.

C. Implementation Details

We implement the proposed HRMNet via the PyTorch (2.0.0) toolbox [65]. Random flipping, cropping, and temporal exchanging are employed on the image patches for data augmentation. The AdamW [66] optimizer is exploited to optimize the proposed HRMNet, setting the weight decay and parameters β_1 and β_2 to 0.01, 0.9, and 0.99, respectively. We adopt the poly learning scheme to adjust the learning rate as $(1 - (\text{cur_iteration}/\text{max_iteration}))^{\text{power}} \times \text{lr}$, setting *power* to 0.9.

For training on the LEVIR+ dataset, we set the batch size and max_iteration to 16 and 5000, respectively. For the MCD dataset, these parameters are also set to 16 for the batch size, with max_iteration adjusted to 20 000. The initial learning rate is established at 0.0005 for all datasets. All experiments are conducted on a single NVIDIA RTX 3090 GPU and Intel Xeon[4] Gold 6330 CPU @ 2.00 GHz, using the Ubuntu 20.04 platform. Each experiment is performed five times, and we report the mean and standard deviation of the results. To ensure the reproducibility of the proposed method, we have released the source code at https://github.com/guanyuezhen/HRMNet

[1] https://github.com/S2Looking/Dataset

[2] http://study.rsgis.whu.edu.cn/pages/download/building_dataset.html

[3] https://rodare.hzdr.de/record/3251

[4] Registered trademark.

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TABLE I

QUANTITATIVE COMPARISONS OF THE PROPOSED METHOD WITH DIVERSE SETTINGS IN TERMS OF κ, IOU, $F1$, REC, AND PRE ON THE LEVIR+ AND BCDD DATASETS. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

No.	Methods	LEVIR+					BCDD				
		κ (%)	IoU (%)	F1 (%)	Rec (%)	Pre (%)	κ (%)	IoU (%)	F1 (%)	Rec (%)	Pre (%)
#1	HRMNet	**84.99±0.23**	**74.81±0.35**	**85.59±0.23**	**84.33±1.57**	86.91±1.19	**76.73±1.49**	**63.30±1.12**	**77.75±1.37**	74.50±1.00	81.31±1.99
(a) Feature Aggregation Module (FAM)											
#2	Cat	**84.99±0.23**	**74.81±0.35**	**85.59±0.23**	**84.33±1.57**	86.91±1.19	**76.73±1.49**	**63.30±1.12**	**77.75±1.37**	74.50±1.00	81.31±1.99
#3	Sum	84.55±1.00	74.66±1.49	85.13±0.97	84.42±0.95	86.87±0.98	76.23±1.03	62.92±1.24	77.24±0.94	**74.93±2.32**	79.86±4.64
(b) Online Hard Region Estimation (OHRE)											
#4	w/o OHRE	84.29±0.01	74.27±0.02	84.89±0.01	83.91±0.18	85.88±0.16	73.61±0.48	59.62±0.52	74.70±0.41	71.09±2.51	78.86±4.00
#5	OHRE w/o UaL	84.09±0.31	73.96±0.45	84.69±0.30	83.57±0.06	84.83±0.55	73.71±1.17	59.74±1.39	74.80±1.09	71.13±0.94	78.93±3.57
#6	OHRE w BS	84.56±0.47	74.40±0.69	85.11±0.45	83.42±0.21	86.59±0.71	74.60±0.59	60.79±0.72	75.62±0.56	69.40±0.05	**83.06±1.28**
(c) Cross-layer Knowledge Review Module (CKRM)											
#7	w/o CKRM	83.39±0.06	72.94±0.08	84.02±0.06	82.52±0.50	85.58±0.41	73.28±1.02	59.95±1.29	74.27±0.97	68.96±1.03	80.88±0.89
#8	CKRM w/o CoA	84.25±0.34	74.19±0.49	84.84±0.33	83.40±0.42	86.33±0.21	74.18±0.79	60.27±0.98	75.20±0.76	68.96±1.10	82.69±0.25
#9	CKRM w/o ReA	84.52±0.52	74.44±0.78	85.14±0.51	82.91±1.18	**87.25±0.21**	74.13±0.63	60.24±0.78	75.18±0.61	70.80±0.62	80.16±0.59
#10	CKRM w/o ReA & CoA	83.64±0.15	73.77±0.23	84.22±0.15	82.82±0.91	86.15±0.66	73.70±0.73	59.97±1.00	74.75±0.77	71.24±5.14	78.97±4.59

D. Ablation Studies

We evaluate the effectiveness of the components and configurations of the proposed HRMNet on the LEVIR+ and BCDD datasets. Each method is performed five times to evaluate the mean and standard deviation of performance measured by κ, IoU, $F1$, Rec, and Pre in Table I.

1) Effectiveness of the FAM: Feature concatenation and summation are two mainstream strategies for aggregating multilevel features. To study their impacts on the feature attention module (FAM), we compare the CD performance of methods using the FAM with feature concatenation (denoted as Cat, #02 in Table I) and summation (denoted as Sum, #03 in Table I). The results show that the FAM with feature concatenation achieves the best performance across most metrics. Therefore, we use concatenation in our FAM.

2) Effectiveness of OHRE: In our proposed HRMNet, OHRE is a key component for formulating the pixel-wise confidence related to the predicted change results and performing hard region learning. To verify its effectiveness, we first remove the OHRE module to set a w/o OHRE method (#04 in Table I), of which the CD performance is reported in Table I. As seen from the results, the CD performance decreases, respectively, about 3.12%, 3.68%, and 3.05% in terms of κ, IoU, and $F1$ on BCDD datasets, indicating the necessity of the OHRE module. In addition to demonstrating the effectiveness of hard region learning in OHRE, we only exploit the multilevel temporal difference features without the features from the OHRE branch to obtain the final CD, which is termed OHRE w/o HRL (#05 in Table I). We can observe that the hard region learning manner can slightly improve the CD performance, verifying its effectiveness.

To achieve OHRE, we constructed a pseudo-hard region supervision by modeling the difference between ground-truth change maps and the final predicted change maps. To verify the effectiveness of the pseudo-hard region supervision, we replaced it with the change boundary supervision and referred to this method as OHRE w BS (#06 in Table I). From the results in Table I, we observed that the pseudo-hard region supervision obtained the best performance. This verifies the effectiveness of the proposed pseudo-hard region supervision.

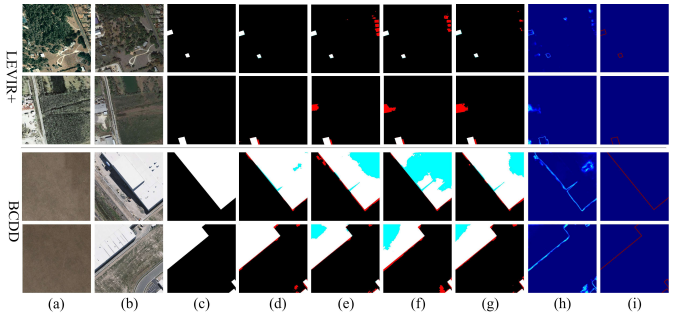


Fig. 4. Visual comparisons of the proposed method and other baselines on the LEVIR+ and BCDD datasets. (a) $\mathbf{I}^1$ images, (b) $\mathbf{I}^2$ images, (c) ground truth, (d) HRMNet, (e) w/o OHRE, (f) OHRE w BS, (g) w/o CKRM, (h) predicted hard region maps, and (i) boundary maps. The rendered colors represent true positives (white), false positives (red), true negatives (black), and false negatives (blue).

3) Effectiveness of the CKRM: We apply the CKRM to distill the fine temporal change knowledge into high-level features and jointly use the multilevel temporal difference feature to generate the final change maps. The CKRM consists of two key attention mechanisms: reverse attention and conflict attention. To validate the effectiveness of the CKRM, we formulate four variants of the proposed HRMNet, denoted as w/o CKRM, CKRM w/o CoA, CKRM w/o ReA, and CKRM w/o CoA and ReA, as #07–#10 in Table I. In w/o CKRM, we remove the CKRM and only use the initial multilevel temporal difference features and hard region-aware features to obtain the final change maps. For the CKRM w/o CoA and CKRM w/o ReA, we separately remove the conflict attention and reverse attention. For the CKRM w/o CoA and ReA, we remove both the conflict attention and reverse attention from the network. From the results in Table I, we observe that using only the reverse attention or conflict attention has worse performance than the default setting. Therefore, the default setting of the CKRM, which uses both reverse attention and conflict attention, is the most effective choice.

To intuitively demonstrate the effectiveness of different components of the proposed method, we present change maps for our approach alongside three baselines: w/o OHRE, OHRE

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TABLE II
QUANTITATIVE COMPARISONS IN TERMS OF κ, IOU, $F1$, REC, AND PRE ON THE LEVIR+, BCDD, AND MCD DATASETS. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

Datasets		FC-diff	FC-ef	FC-cat	ChangeStar	BIT	MSCANet	TFI-GR	A2Net	UCSFH	SEIFNet	HRMNet
Years		2018	2018	2018	2021	2022	2022	2022	2023	2024	2024	-
MACs (G)		18.74	14.19	21.16	53.37	32.45	58.19	41.30	11.87	11.42	32.94	48.35
Params (M)		1.35	1.35	1.55	13.28	3.50	17.11	27.88	3.78	4.81	28.60	13.46
LEVIR+	κ (%)	64.68±1.40	33.77±1.77	57.78±0.11	82.79±1.02	82.94±0.88	82.81±0.02	81.86±0.71	82.14±0.49	83.19±0.69	81.84±1.06	**84.99±0.23**
	IoU (%)	49.33±1.48	22.26±1.27	42.42±0.10	71.66±1.44	71.88±1.24	71.66±0.03	70.36±0.97	70.75±0.69	72.22±0.98	70.31±1.49	**74.81±0.35**
	F1 (%)	66.06±1.33	36.41±1.70	59.57±0.10	83.49±0.98	83.64±0.84	83.50±0.02	82.61±0.67	82.87±0.47	83.86±0.66	82.56±1.03	**85.59±0.23**
	Rec (%)	63.37±0.52	35.71±1.58	61.79±0.28	82.90±1.04	83.91±0.37	80.38±0.08	83.04±0.11	83.53±0.18	83.28±1.44	80.96±1.99	**84.33±1.57**
	Pre (%)	69.00±2.28	37.13±1.82	57.50±0.42	84.09±0.91	83.36±1.31	86.85±0.05	82.18±1.21	82.22±0.75	84.47±0.13	84.25±0.01	**86.91±1.19**
BCDD	κ (%)	45.53±0.49	17.41±7.21	37.78±4.00	73.49±1.67	69.18±1.08	63.02±0.14	75.38±1.99	74.43±0.13	75.63±0.88	69.76±0.82	**76.73±1.49**
	IoU (%)	30.73±0.37	10.73±4.38	24.64±3.13	59.59±1.99	54.29±1.26	47.48±0.10	62.14±2.52	60.66±0.11	62.13±1.13	55.02±1.07	**63.30±1.12**
	F1 (%)	47.01±0.44	19.25±7.15	39.48±4.03	74.67±1.56	70.37±1.06	64.39±0.08	76.39±1.90	75.51±0.09	76.64±0.86	70.98±0.88	**77.75±1.37**
	Rec (%)	34.22±0.12	12.98±5.04	28.86±4.04	**76.18±0.63**	63.36±1.83	55.91±1.46	75.61±1.59	73.66±2.31	72.87±1.75	66.81±5.79	74.50±1.00
	Pre (%)	75.17±2.85	37.22±12.01	62.75±1.24	73.28±3.59	79.13±0.18	76.00±2.94	78.20±2.22	77.56±2.75	80.84±0.25	76.22±5.50	**81.31±1.99**
MCD	κ (%)	36.22±1.24	41.81±0.81	33.32±1.49	66.72±0.12	61.25±0.66	63.19±0.16	64.88±1.44	66.99±0.01	67.25±1.16	67.91±1.23	**68.25±1.30**
	IoU (%)	26.45±0.13	30.89±0.60	24.81±0.39	53.38±1.20	47.58±0.65	49.86±1.37	51.47±0.27	54.16±0.40	54.20±0.22	54.92±0.20	**55.30±1.30**
	F1 (%)	41.83±0.87	47.19±0.82	39.76±0.22	69.60±1.28	64.48±0.93	66.54±0.53	67.96±0.77	69.55±0.60	70.30±0.11	70.90±0.36	**71.22±0.18**
	Rec (%)	40.40±0.28	47.87±0.36	42.19±0.63	66.35±0.07	59.40±1.35	66.42±1.42	**83.31±0.74**	67.08±0.73	71.85±0.51	72.58±1.35	73.04±0.55
	Pre (%)	43.36±0.17	46.54±1.17	37.58±0.58	**73.19±0.36**	70.52±0.61	66.66±0.14	57.39±0.20	72.19±1.41	68.81±1.43	69.30±0.86	69.48±0.09

w BS, and w/o CKRM, as shown in Fig. 4. The results indicate that the proposed HRMNet outperforms the three baselines in identifying both irrelevant pseudo-changes and overall changes. In Fig. 4, the boundary maps concentrate solely on the edges of changed regions, while the predicted hard region maps effectively highlight both changed and unchanged areas. CD results with pseudo-hard region supervision exceed those using change boundary supervision, confirming the effectiveness of the pseudo-hard region supervision. Additionally, detection errors in the w/o CKRM condition are lower than those in the w/o OHRE condition, suggesting that OHRE is more effective than cross-layer knowledge review in suppressing irrelevant pseudo-changes. These findings emphasize the importance of both OHRE and cross-layer knowledge review in the proposed method.

E. Comparison With the State-of-the-Art

We compare the proposed model with ten state-of-the-art remote sensing CD approaches, including FC-Diff [67], FC-ef [67], FC-Cat [67], ChangeStar [24], BIT [58], MSCANet [68], TFI-GR [34], A2Net [59], UCSFH [69], and SEIFNet [70]. We reproduce the CD results by using their released code under their default parameters for fair comparisons. Each method is performed five times to evaluate the mean and standard deviation of performance measured by κ, IoU, $F1$, Rec, and Pre in Table I.

1) Quantitative Evaluation: We present the quantitative evaluation results of various methods for remote sensing CD, measured by κ, IoU, $F1$, Rec, and Pre, in Table II. From the results, we observe that the proposed method outperforms other competitors in most cases. For instance, on the LEVIR+ dataset, the proposed HRMNet achieves approximately 1.80%, 2.59%, and 1.73% higher performance improvements in terms of κ, IoU, and $F1$, respectively, than the second-best results. On the BCDD dataset, the proposed HRMNet achieves around 1.10, 1.16, and 1.11 percentage higher performance improvements in terms of κ, IoU, and $F1$, respectively, than the second-best results. On the MCD dataset, the proposed HRMNet exceeds about 0.34%, 0.38%, and 0.32% higher performance in terms of κ, IoU, and $F1$, respectively, than the second-best results. In addition, Table II provides the model parameters (Params) and computation costs measured by multiply–accumulate operations (MACs). The results show that the proposed HRMNet has comparable model parameters and computation costs to some heavy models, such as the SEIFNet, TFI-GR, ChangeStar, and MSCANet. These findings strongly demonstrate the effectiveness and superiority of the proposed HRMNet.

2) Qualitative Evaluation: We present visual comparisons of various methods on the LEVIR+, BCDD, and MCD datasets in Fig. 5. White, red, black, and blue are used to indicate true positives, false positives, true negatives, and false negatives, respectively, for better visualization. Based on the visual results, we observe that the proposed method demonstrates superiority in the following aspects.

1) *Advantages in Content Integrity*: In Fig. 5, it can be observed that some of the compared methods, such as ChangeStar, BIT, MSCANet, TFI-GR, A2Net, and SEIFNet, fail to detect the entire changed regions. On the other hand, the proposed HRMNet can accurately identify the changed objects with fine content integrity. The superior performance of our approach is attributed to two main reasons. First, we employ a knowledge review approach to distill the fine-grained change information into coarse layers, and we jointly utilize the multilevel change information to comprehensively generate accurate change results. Second, our method estimates the hard regions, and we use a hard region learning approach to encourage the network to focus more on such difficult regions, resulting in more precise CD results.
2) *Advantages in Location Accuracy*: The visual results in Fig. 5 demonstrate that the proposed HRMNet accurately identifies the changed regions and effectively eliminates the irrelevant pseudo-changes caused by background clutter in bi-temporal images. In contrast, several other methods, such as ChangeStar, BIT,

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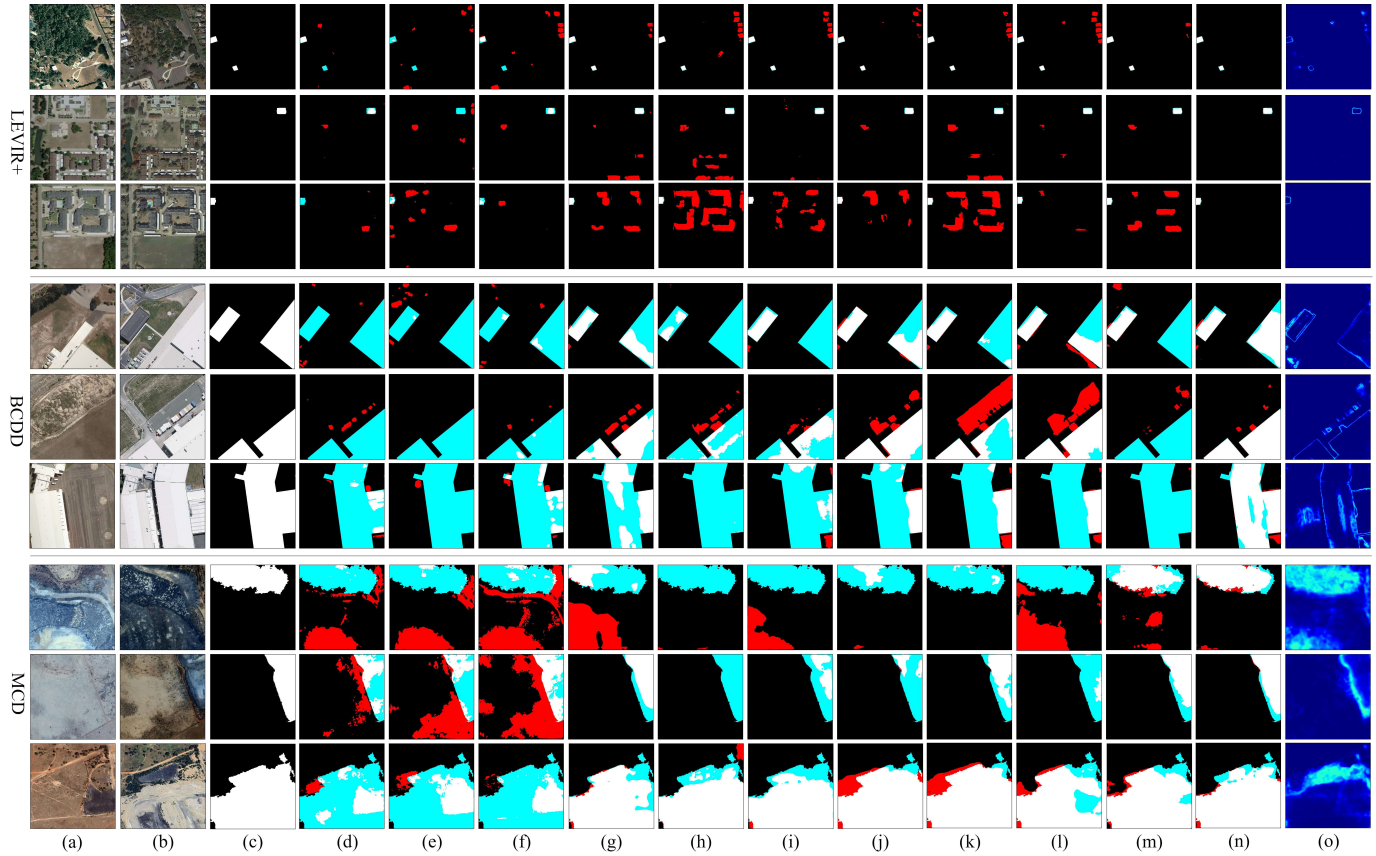


Fig. 5. Visual comparisons of the proposed method and the state-of-the-art approaches on the three datasets. (a) $\mathbf{I}^1$ images, (b) $\mathbf{I}^2$ images, (c) ground truth, (d) FC-diff, (e) FC-ef, (f) FC-cat, (g) ChangeStar, (h) BIT, (i) MSCANet, (j) TFI-GR, (k) A2Net, (l) UCSFH, (m) SEIFNet, (n) HRMNet, and (o) predicted hard region maps. The rendered colors represent true positives (white), false positives (red), true negatives (black), and false negatives (blue).

MSCANet, TFI-GR, A2Net, and SEIFNet, fail to eliminate the irrelevant pseudo-changes. These results underscore the efficacy of the proposed HRMNet.

3) *Advantages in Detecting Changes in Hard Regions*: As shown in Fig. 5, the proposed HRMNet performs well on some hard regions, including the change boundaries of large-scale changes, tiny changed buildings, and irrelevant pseudo-changes. In addition, the proposed HRMNet incorporates a hard region estimation approach to provide pixel-wise confidence levels associated with the predicted change maps. Our results in Fig. 5 indicate that the estimated hard region maps are mainly concentrated along the hard regions in the bi-temporal images that other methods often cannot accurately identify. Besides, when the proposed method predicts a relatively correct CD result, the estimated hard region maps are mainly concentrated along the boundaries of changes, indicating high confidence when the predicted change maps are accurate. This highlights the effectiveness of the proposed hard region estimation approach.

V. Conclusion

In this work, we present a novel CD network designed to deliver accurate results in challenging regions, including boundaries of changes, small changes, and irrelevant pseudo-changes. Our proposed method introduces a CKRM and an OHRE branch, both of which significantly enhance CD performance, particularly in difficult cases, guided by a hard region mining strategy. Experimental results from three high-spatial-resolution remote sensing CD datasets demonstrate that our approach surpasses existing state-of-the-art methods.

The proposed method is primarily based on the pretrained ResNet. It is important to note that large vision–language models [71], [72] can be leveraged to enhance the feature representation capability of CD methods by utilizing multimodality information, thereby achieving more accurate CD performance. Furthermore, training CD models with less expensive weak annotations has emerged as a promising direction in remote sensing CD [73]. In the future, we aim to design CD models based on vision–language models within a weakly supervised learning paradigm to effectively address challenging detected changes.

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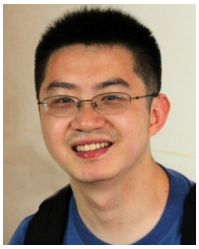


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